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SUPERSEDING
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MILITARY SPECIFICATION

PROTRACTORS, FAN., RANGE DEFLECTION: ALUMINUM. GRADUATED IN MILS AND METERS

This Specification has been approued by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.–This specification covers aluminum range deflection protractor graduated in mils with a fixed arm graduated in meters.

1.2 Classification~Range deflection protractors shall be of the following classes, as specified (see 6.2) :

Class l–-12,000-meter range; 1:25,000 scale.
Class 2–-15,000-meter range; 1:25,000 scale.
Class 3--25,000-meter range; 1:25,000 scale.
Class 4--50,000-meter range; 1:50,000 scale.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS
FEDERAL
QQ-A-327–Aluminum AlloY Plate and Sheet 6061.
PPP-B-566-Boxes, **Folding, Paperboard.**
PPP-B-585-Boxes; Wood, Wirebound.
PPP-B-601-Boxes, Wood, Cleated-Plywood.
PPP-B-621 –Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636–Box, Fiberboard.
MLITARY
MIL-A-8625--Anodic Coatings, for Aluminum and Aluminum Alloys.

STANDARDS
MILITARY
MIL-STD-105--Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129–-Marking for Shipment and Storage.
MIL-STD-130--Identification Marking of U. S. Military Property.

DRAWINGS
CORPS OF ENGINEERS
D 8483–l-Protractor, Range Deflection, Aluminum, Graduated in Mils and Meters, Scale 1:25,000 and 1:50,000.

{Copies of specifications. standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

FSC 1290

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(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street NW, Washington, D. C.)

3. REQUIREMENTS

3.1 Fabrication. The range deflection protractors shall be fabricated as detailed in the following drawing part numbers and as specified herein.

Protractor	Drawing part number
Class 1	D 8483-1-1
Class 2	D 8483-1-2
Class 3	D 8483-1-3
Class 4	D 8483-1-4

32 The drawing forming a part of this specification is an engineering design drawing and is not intended for use as a shop drawing. Critical dimensions and tolerances are furnished; however, the contractor is responsible for preparing his own show drawings. Where tolerances are not shown or *where* those prescribed could cumulatively result in improper fits, the contractor is responsible for providing tolerances in accordance with industry standards for similar equipment and within those prescribed on the engineering design drawing to insure proper fit, assembly, and operation of the protractors. No deviation from dimensions or tolerances shown on the engineering design drawing is permissible without prior approval of the contracting officer.

3.3 Preproduction model. The preproduction model is defined as a production model which will be identical to the protractors which the manufacturer will subsequently produce in fulfillment of the contract. As soon as practicable after award of the contract or order and prior to submission of any protractors for final acceptance, the contractor shall furnish a preproduction model for examination and test to determine conformance to this specification. Examination and tests shall be those specified herein. Approval of the preproduction model by the activity concerned shall not relieve the cox~-tractor of his obligation to supply protractors conforming to this specification. Any changes or deviations of protractors from the preproduction model shall be subject to the approval of the contracting officer.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawing.

3.4.1 *Aluminum alloy sheets.* Aluminum alloy sheets shall conform to Specification QQ-A+327, temper T651.

3.5 Markings. The graduations, numerals, and other markings shall be etched or scribed on the surface of the protractor and then shall be filled with a permanent black medium.

3.6 Straightness. The ruling edge of protractor shall be straight and shall ensure that deflection angles measured from any point along the ruling edge are accurate to within plus or minus 0.5 roil.

3.7 Finishing. The protractors shall be anodized before the addition of permanent black medium to the markings in accordance with Specification MIL-A-8625, type IX, nondyed. The protractors shall have a smooth, natural-aluminum, satin finish.

3.8 Identification marking. Each protractor shall be identified in accordance with Standard MIL-STD-130 and as shown on the drawing.

3.9 Workmanship. The protractors shall be free from visible scratches, dents, flaws, or other imperfections. The edges shall have a smoothness equivalent to that obtained by milling and subsequent polishing. Punched or sheared edges are not acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable t-o the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contractor order.

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The Government reserves the right to Perform any of the inspections set forth in the specification where such instructions are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection Inspection shall be classified as follows:

(a) Preproduction inspection of the protractors.
(b) Production inspection of the protractors.
(c) Inspection of preparation for delivery.

4.3 Preproduction inspection.

4.3.1 *Examination. The* protractor shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

101. Dimensional nonconformance to the drawings.
102. Nicks, burrs, or rough edges on ruling and protractor edge or face.
103. Graduation marks not sharply defined, missing, or not of proper width.
104. Graduation marks not filled as specified. Black medium chipped or peeling.
105. Identification marking not as specified.

4.3.2 Tests. The protractor shall be tested as specified in table I. Failure to pass any test shall be cause for rejection.

4.4 Production inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with Standard MIL-STD-105. Inspection level shall be L-II for examination and L-1 for tests.

4.4.2 *Examination.* Samples selected in accordance with 4.4.1 shall be examined for the defects listed in 4.3.1. AQL shall be 2.5 percent defective.

4.4.3 *Tests.*. Samples selected in accordance with 4.4.1 shall be tested as specified in table I. AQL shall be 0.10 percent defective. *Any* sample failing to pass one or more tests shall be considered defective.

4.5 Inspection schedule Inspection schedule is contained in table I.

Type of Inspection		Examination or test	Inspection paragraph	Requirements paragraph
Preproduction	Production sample			
x	x	Examination	4.3.1	3.1, 3.2, 3.4, 3.5, 3.7, and 3.8
x	x	Graduation accuracy	4.6.1	3.5
x	x	Ruling edge straightness	4.6.2	3.6

• Perform inspection marked x.

4.6 Test procedures.

4.6.1 *Linear and angular graduation accuracy test.* The accuracy of linear graduations shall be determined by direct comparison with a master scale which is accurate to within plus or minus 0.001 inch at all points or by means of a beam compass and meter bar having a least reading of not more than 0.025 millimeter. The accuracy of angular graduations shall be determined by direct comparison with a master scale which is accurate to within plus or minus 0.1 mil or by means of a circular measuring machine, sine bars, or angular gage blocks of similar precision. Evidence of any graduation line, either linear or angular, not within the tolerances specified in 3.5 shall constitute failure of this test.

4.6.2 *Ruling edge straightness test.* The straightness of the ruling edge of the range

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arm shall be determined by drawing two straight lines not less than 48 inches long and not more than 0.005 inch wide on a sheet of paper to form an angle of 1 mil plus or minus 0.1 roil. A pin with a diameter between 0.040 and 0.041 inch shall be positioned at the point of intersection of the two lines. The angle formed by the two lines shall be bisected by a straight line not more than 0.003 inch wide and extending from a point within 6 inches of the pin to the outer ends of the intersecting lines. The protractor pivot point shall be firmly seated on the pin during the test. The protractor shall be rotated about the pin until the ruling edge contacts the line bisecting the angle formed by the test lines at the outermost graduation of the range arm. Evidence of any deviation from straightness which causes the ruling edge of the range arm to lie outside either of the two test lines at *any* point between the numbered graduations shall constitute failure of this test.

4.7 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined for compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* The protractors shall be individually packaged in commercial containers or in boxes conforming to Specification PPP-B-566.

5.1.1.1 *Intermediate packaging.* The protractors of like size and description shall be intermediate packaged together in quantities not to exceed 12 in a box conforming to Specification PPP-B-636, class 2, compliance symbol as applicable, style optional.

5.1.2 *Level C. The* protractors shall be packaged to afford protection against deterioration and damage.

5.2 Packing. Packing shall be level A, 33, or C, as specified (see 6.2).

5.2.1 Level *A.* The intermediate packaged protractors of like size and description shall be packed in boxes conforming to Specification PPP-B-621, class 2, style optional, or in boxes conforming to Specification PPP- B-601, overseas type style optional. The boxes shall be close fitting and strapping shall be zinc coated. The gross weight of each box shall not exceed 200 pounds. -

5.2.2 *Level B. The* protractors of like size and description shall be packed in boxes conforming to Specification PPP-B-621, Class 1, style optional, in boxes conforming to Specification PPP-B-601, domestic type, style optional, or in boxes conforming to Specification PPP-E-585, class 1 or 2, style optional. The gross weight of each box shall not exceed 200 pounds.

5.2.3 *Level C.* The protractors shall be packed to insure safe arrival at destination at lowest, rates in containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

5.3 Marking. Packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The protractors are intended for use in graphical determination *of* artillery ranges and deflection angles to targets from gun positions and reference points plotted on 1:25,000 *or* 1:50,000 scale firing charts.

6.2 Ordering data Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of protractor required (see 1.2).
(c) Administrative provisions for inspection records (see 4.1).
(d) Level of packaging and level of packing required (see 5.1 and 5.2).

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